

Loops

Bit by Bit Coding - Term 2 | Taster 3

Key Concepts

- while loop repeats while a condition is True.
- for loop iterates over a sequence (range, list, string).
- range(start, stop, step) makes numbers - stop is NOT included.
- break exits the loop immediately; continue skips to the next round.
- Loops can be nested - a loop inside a loop.

Syntax Reference

for loop & range()

```
for i in range(5):      # 0,1,2,3,4
    print(i)
for i in range(2, 6):   # 2,3,4,5
    print(i)
for i in range(0, 10, 2): # 0,2,4,6,8 (step)
    print(i)
```

while loop

```
count = 0
while count < 3:
    print(count)
    count += 1      # remember to update!
```

break & continue

```
for i in range(10):
    if i == 5:
        break      # stops the whole loop
    if i == 2:
        continue   # skips this round only
    print(i)
```

Nested Loops

```
for row in range(3):
    for col in range(3):
        print(f"({row},{col})", end=" ")
    print()      # new line after each row
```

Common Patterns

- Loop over a string: for ch in "Hi":.
- Accumulate a total: total = total + num inside the loop.
- Build a grid or pattern using nested loops.

Tips & Common Mistakes

- Infinite loop: forgetting to update the while-condition variable.
- range(5) gives 0..4, not 1..5.
- break stops the loop; continue only skips one iteration.

- In a while loop, track your position with a separate counter variable.